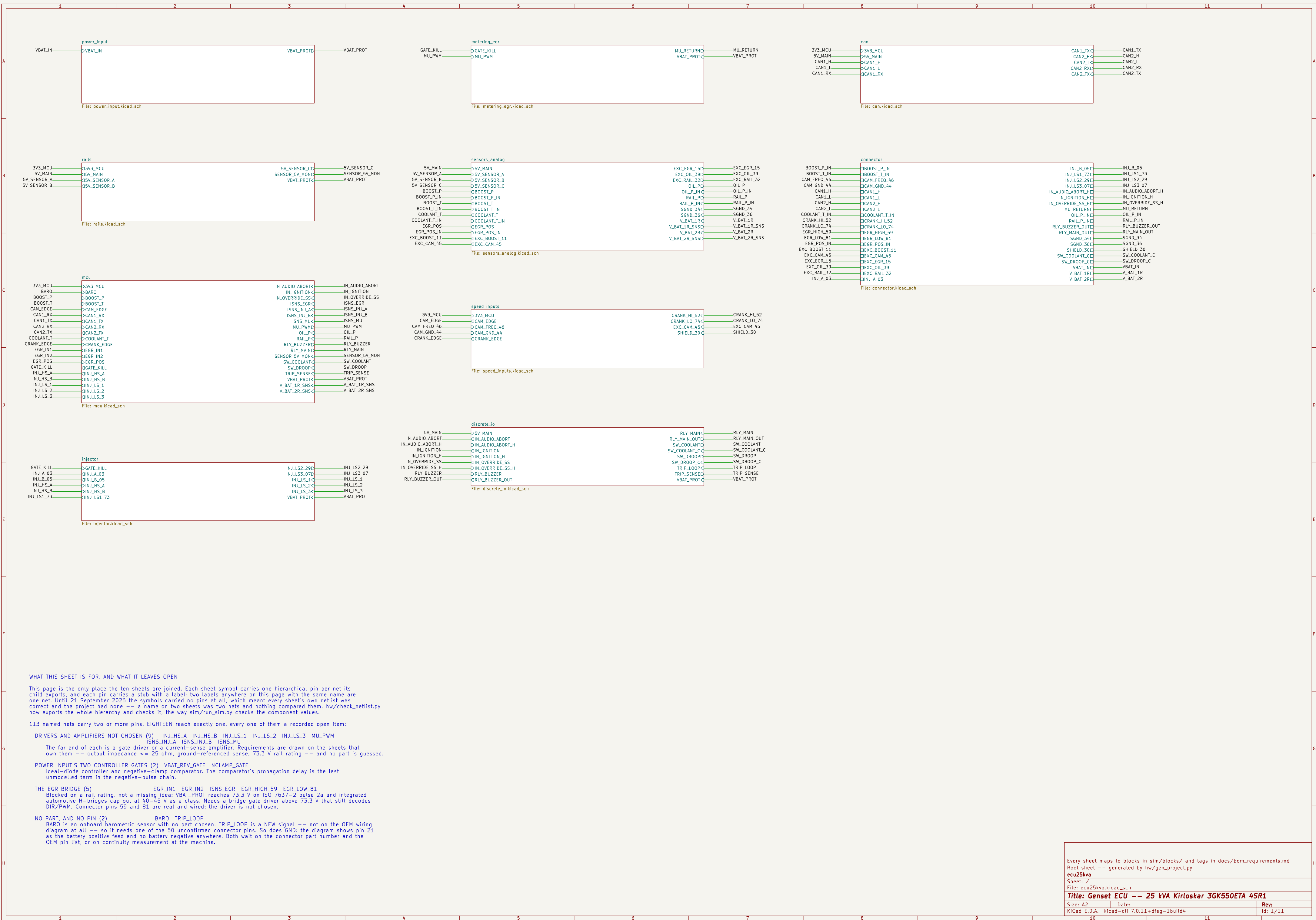
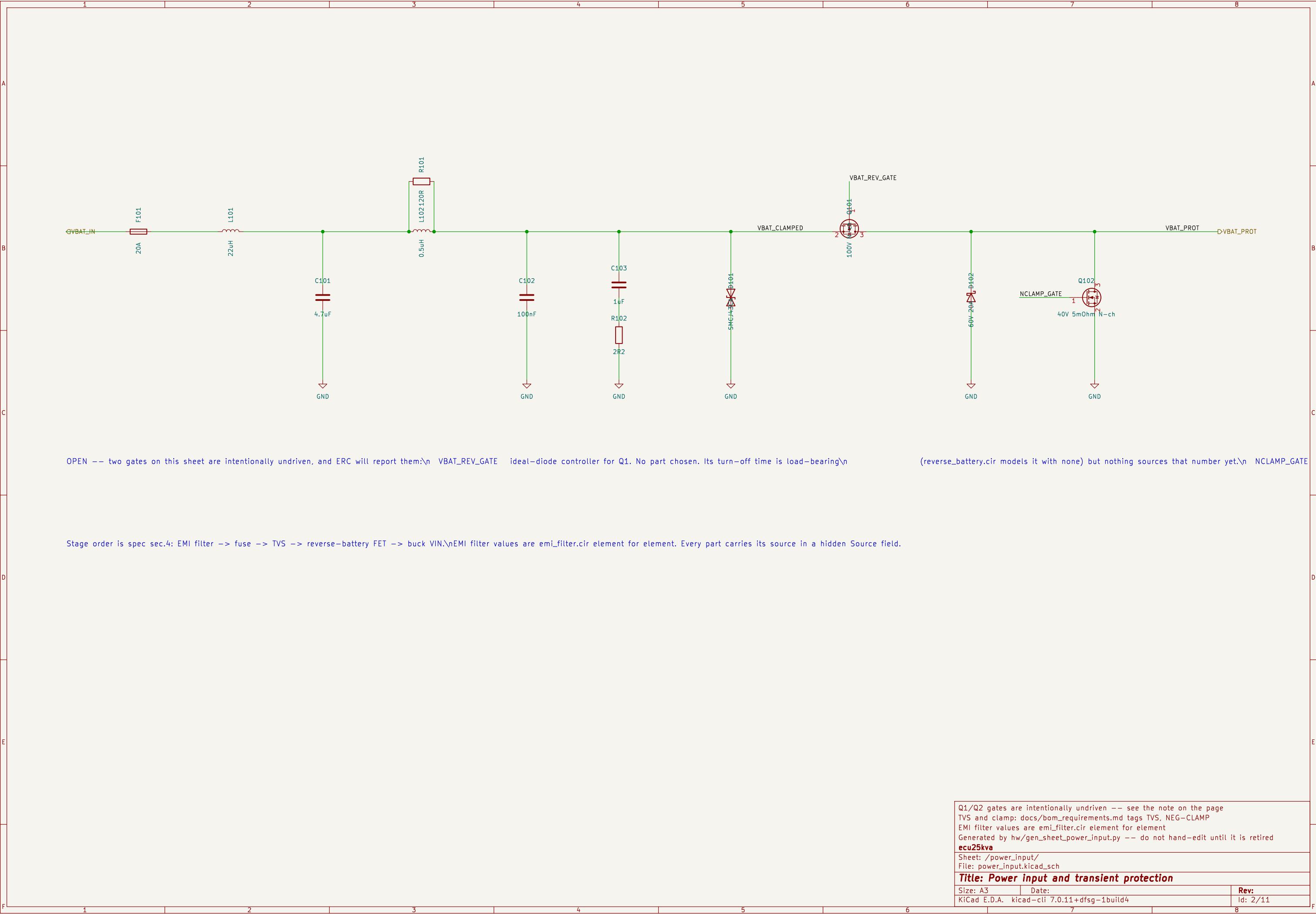






OPEN -- two gates on this sheet are intentionally undriven, and ERC will report them:\n VBAT_REV_GATE ideal-diode controller for Q1. No part chosen. Its turn-off time is load-bearing\n (reverse_battery.cir models it with none) but nothing sources that number yet.\n NCLAMP_GATE comp

Stage order is spec sec.4: EMI filter -> fuse -> TVS -> reverse-battery FET -> buck VIN.\nEMI filter values are emi_filter.cir element for element. Every part carries its source in a hidden Source field.

Q1/Q2 gates are intentionally undriven -- see the note on the page

TVS and clamp: docs/bom_requirements.md tags TVS, NEG-CLAMP

EMI filter values are emi_filter.cir element for element

Generated by hw/gen_sheet_power_input.py -- do not hand-edit until it is retired

ecu25kva

Sheet: /power_input/

File: power_input.kicad_sch

Title: Power input and transient protection

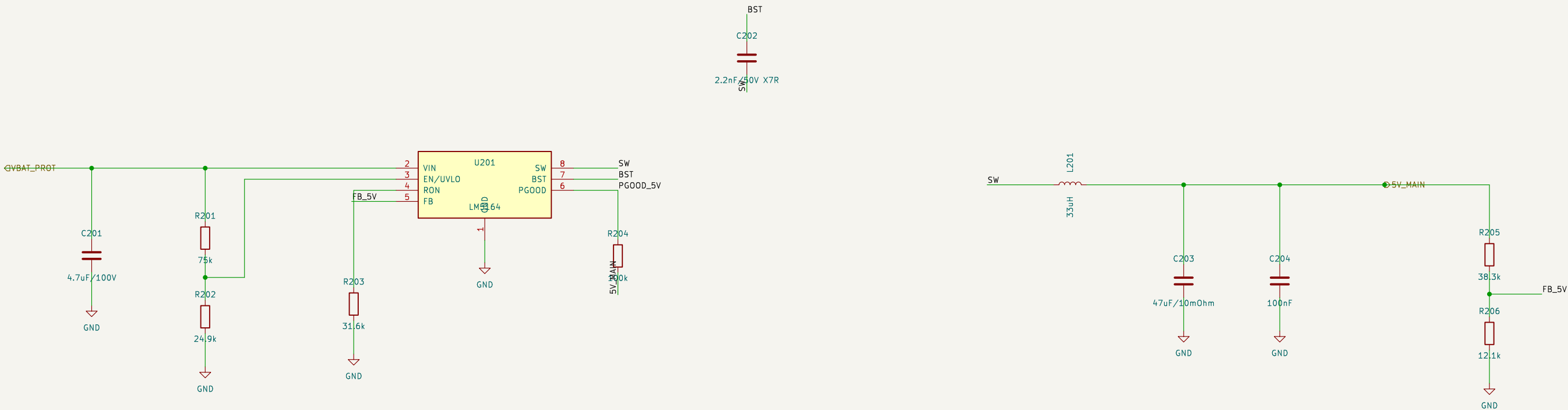
Size: A3

Date:

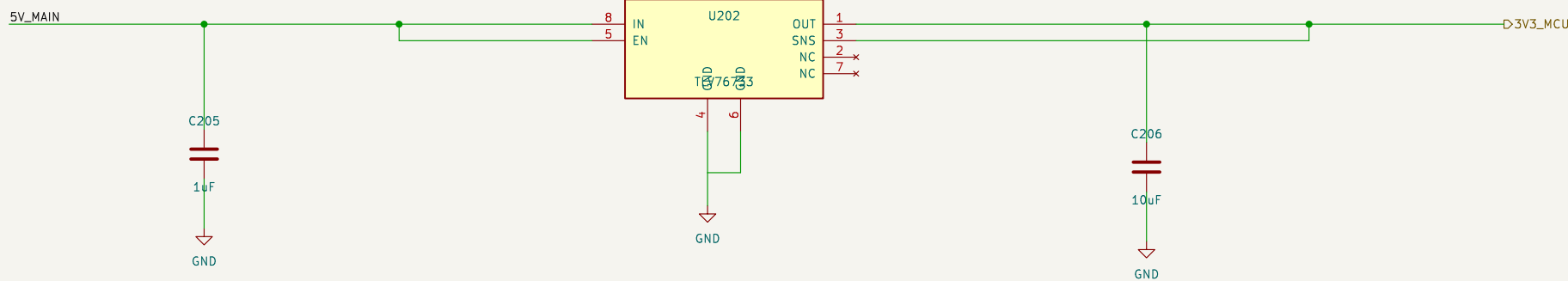
Rev:

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

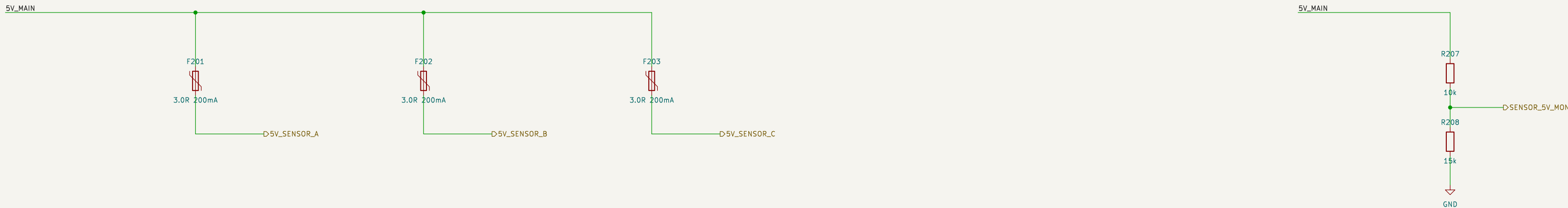
Id: 2/11



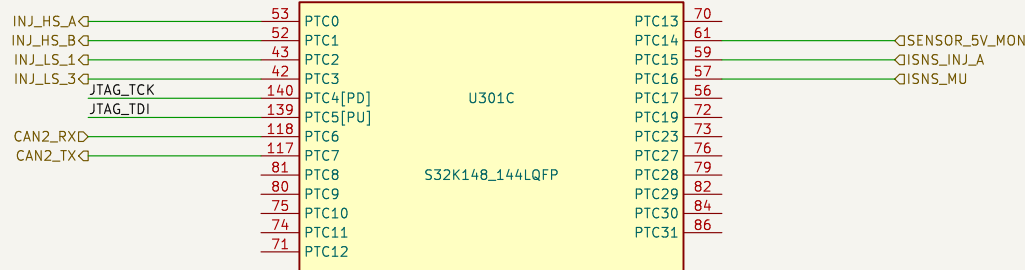
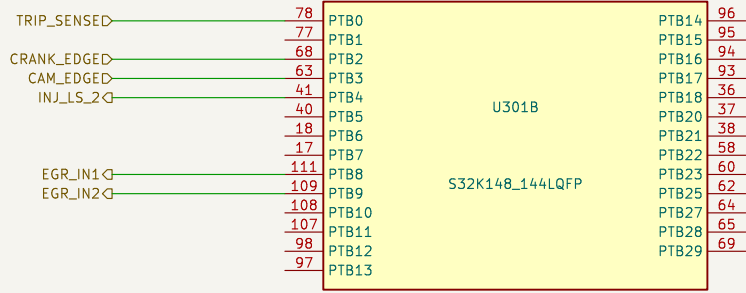
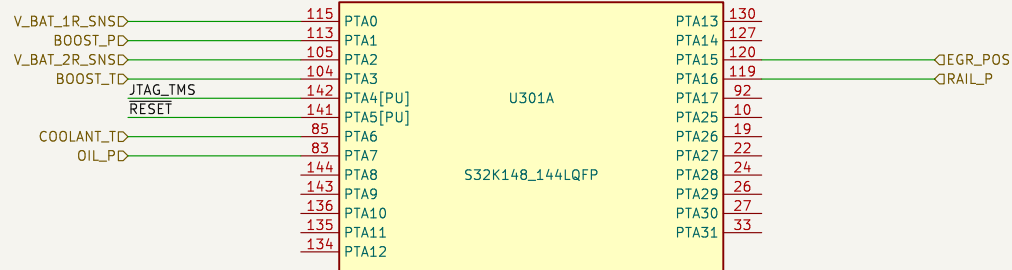
TWO PARTS CHOSEN HERE, both replacing research memo 07 entries that were marked representative rather than chosen:
LM5164 memo 07 sec.4 names a 60 V TPS54360B-Q1. transient_clamp.cir now carries a PASSING check that reads "60 V-class parts are NO LONGER viable downstream" -- raising the TVS standoff to 43 V, so the part stops conducting during a NORMAL clamped dump, put the pulse 2a clamp at 73.3 V. Every rating check in the suite has been written against the LM5164's own -0.3 V / 100 V VIN limits all along; the BOM was simply never reconciled with it. A 60 V buck here is a part that fails the first pulse 2a event.
TLV76733 memo 07 sec.6 names TLV1117-33QDCYRQ1, explicitly "class pricing, not fetched". The supervisor chosen on the mcu sheet made 3V3 accuracy load-bearing: outside 3.143-3.459 V the board resets itself. TLV767-Q1 is 1% over load AND temperature, giving 3.267-3.333 V.



RATIOMETRIC MONITOR, a limitation worth stating: the divider taps 5V_MAIN, upstream of the PTCs. Each group's own PTC drops about 50 mA x 3.0 ohm = 150 mV, which is 3% of the rail and is NOT captured by a measurement taken before it. Correct for the regulator's drift, not for the protection element's. Moving the tap downstream would fix one group and leave the other two wrong, so the tap stays common and the limit is recorded here.



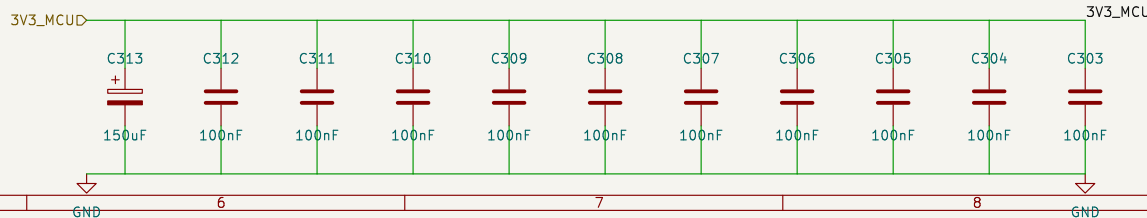
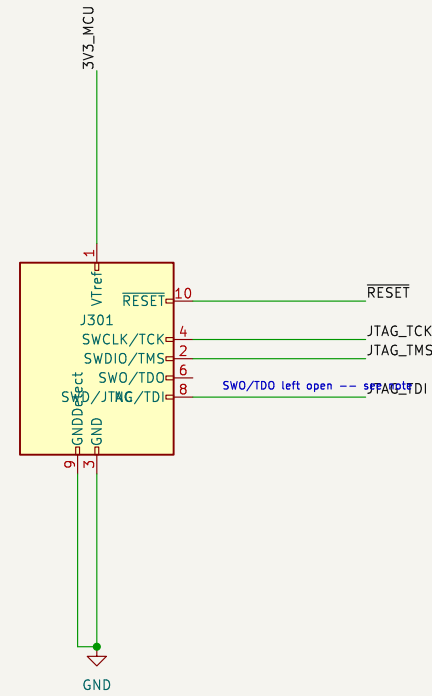
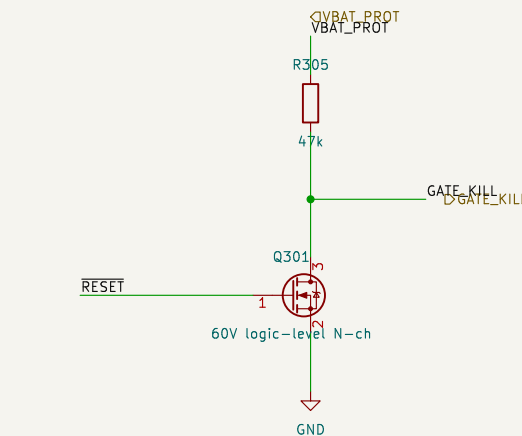
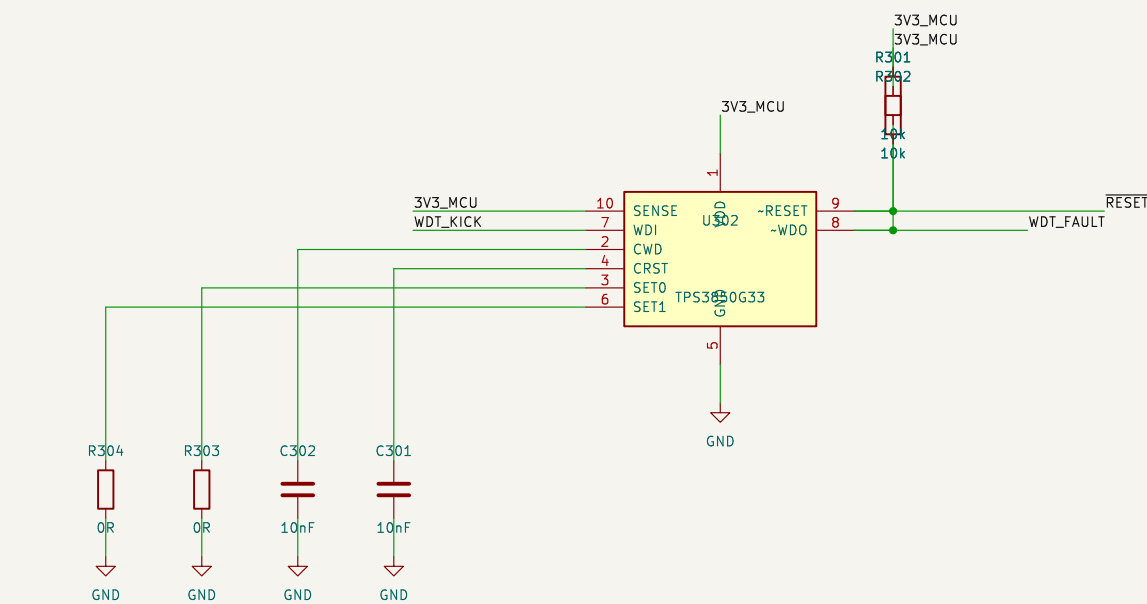
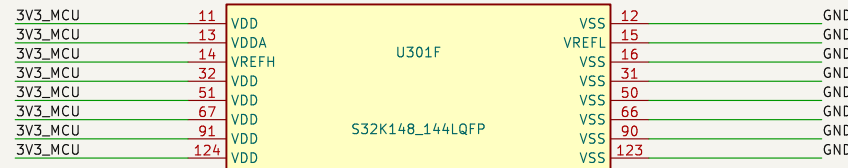
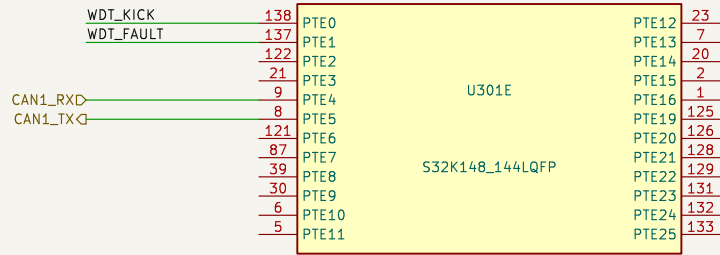
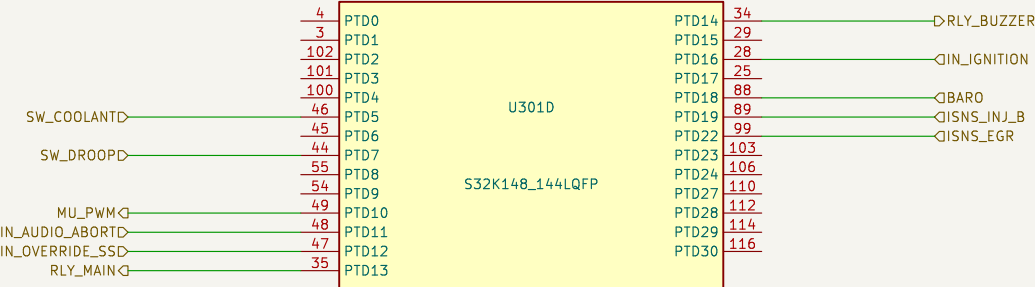
PGOOD_5V is brought out and left unassigned. The LM5164 asserts it open-drain; the pin map has no channel for it. Available, not wired to anything -- shown rather than hidden.



PIN MAP ADDITIONS MADE ON THIS SHEET -- both are gaps in docs/pinmap.md, not changes to it:
PTE0 WDT_KICK MCU -> TPS3850 WDI, pinmap.md assigns 37 signals and none of them services the external watchdog, but memo 11's topology requires the MCU to kick it. A window watchdog nobody feeds resets the board on a timer.
PTE1 WDT_FAULT TPS3850 WDO -> MCU, so firmware can tell a watchdog trip from a rail trip.
Both are free GPIO-HD pins, PE=0/PS=0 at reset, from the 91 the pin map leaves spare.

SUPERVISOR CHOICE -- TPS3850G33, from the datasheet's own nomenclature table (G = thresholds at +/-4% of nominal, H = +/-7%) and its +/-0.8% accuracy spec:
G33 UV 3.168 V nom, 3.143 V worst-low | H33 UV 3.069 V nom, 3.044 V worst-low
The supervisor has to assert BEFORE the MCU's own LVD (3.0 V max) and above the 2.97 V correctness floor, G33 clears LVD by 143 mV worst case; H33 by 44 mV, close enough that the two could fire in either order. The cost is that the rail must stay inside 3.143-3.459 V -- a +/-2% regulator spans 3.234-3.366 V, leaving -90 mV each end. That is this choice's constraint on the rails sheet.

OPEN: SET0/SET1 strap the watchdog window ratio, and the right ratio depends on a firmware loop period that does not exist yet (phase 4). Link resistors, not hard wires.

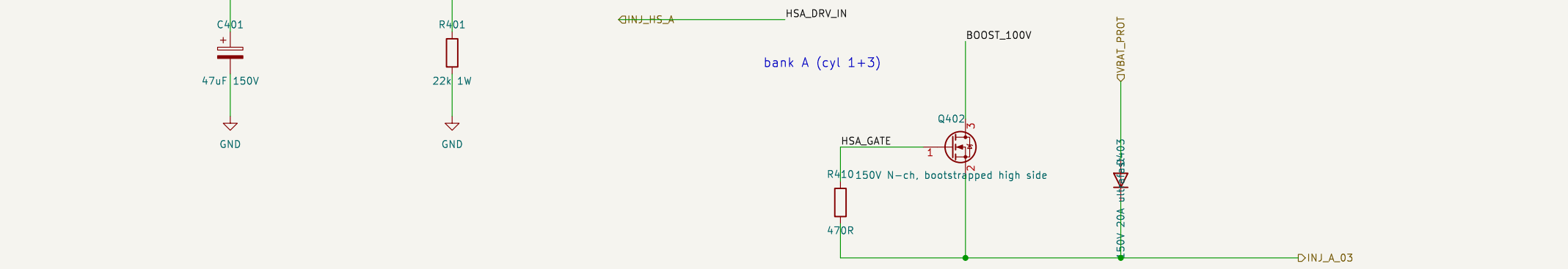


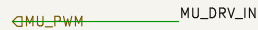
PTE0/PTE1 are added here -- pinmap.md assigns no watchdog pins
Supervisor TPS3850G33: memo 11, sec.2.1 + TI SBVS301B Table 5-1
Pin assignments: docs/pinmap.md. Pin positions: hw/lib/s32k148_layout.json
Generated by hw/gen_sheet_mcu.py -- do not hand-edit until it is retired
ecu25kva
Sheet: /mcu/
File: mcu.kicad_sch
Title: MCU, supervisor and debug
Size: A2 Date: Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 4/11

BOOST RAIL -- 100 V reservoir for the injector peak phase

BOOST_100V

the converter is NOT drawn -- it sits between VBAT_PROT and BOOST_100V; see the note at the foot of the sheet





IN5S_MU_SENSE goes to a current-sense amplifier, also not chosen. 50 mOhm at the 0.675 A setpoint is 33.75 mV, measured ACROSS the shunt -- the amplifier's low reference is the shunt's own ground pad, which is electrically GND and so is not a separate net. It IS a layout constraint: a Kelvin tap at the pad, not a via into the ground pour, because 33.75 mV sits on a return carrying up to 1.35 A and a few milliohms of pour between pad and reference is the whole signal.

Output goes to PTC16 (ADCO_SE14). The loop is closed in firmware: PTD10 carries the PWM, PTC16 reads the current back. That is arrangement the pin map already assumes, and it is why metering_unit_pwm.cir models the regulator as hysteretic rather than as a fixed duty.

T1 DRV8873-Q1 was the candidate and it fits the functional requirement exactly: an automotive H-bridge with the PH/EN decoded interface bus requirements; EGR-DRIVER tag asks for -- one input pair that cannot express both legs of a diagonal conducting at once, which is the requirement egr_hbridge.cir's 270 A shoot-through produced. AEC-Q100 grade 1, 10 A peak, per-half-bridge current sense on IPROPI1/IPROPI2 that would have fed PDD22 directly.

Its VM absolute maximum is 40 V (SLVSDY7B sec.6.1).

VBAT_PROT reaches 40.0 V for 400 ms on a suppressed load dump and 73.3 V for about 50 us on ISO 7637-2 pulse 2a. An absolute maximum is absolute: out of spec at the dump with zero margin, and at 2a by 1.8x.

THIS IS A FACT ABOUT THE RAIL, NOT ABOUT ONE PART. It applies to the injector high sides, the relay drivers and this bridge alike. transient_clamp.c now carries it as enforced checks rather than as a sentence in a comment, so the next driver sheet fails the suite instead of rediscovering it.

Consequence: integrated automotive H-bridges and smart switches cap out at 40–45 V as a class and are excluded from anything battery-connected here. The EGR bridge goes discrete, for the same reason memo 12 sent the injector drivers discrete. What it needs is a bridge GATE DRIVER rated above 73.3 V that still decodes DIR/PWM internally. No such part has been checked against a retrieved datasheet, and this project does not guess parts into copper.

Unblocked by: one datasheet. The four bridge FETs, the shunt, the gate pulldowns and the kill transistors are all already specified and will draw the same way the metering unit's do on the left of this sheet.

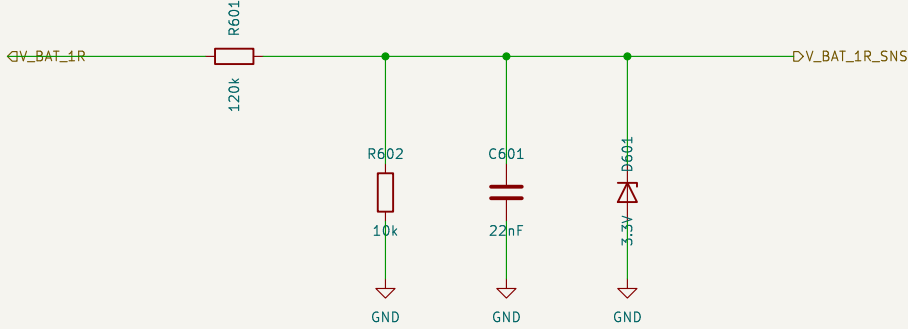
Nothing battery-connected on this board may be rated below 80 V
 EGR half NOT drawn -- blocked on a battery-rail rating, see the note
 Metering unit: metering_unit_pwm.cir + bom_requirements MU=ISENSE, GATE=PULLDOWN
 Generated by hw/gen_sheet_metering_egr.py -- do not hand-edit until it is retired

Sheet: /metering_egr/
File: metering_egr.kicad_sch

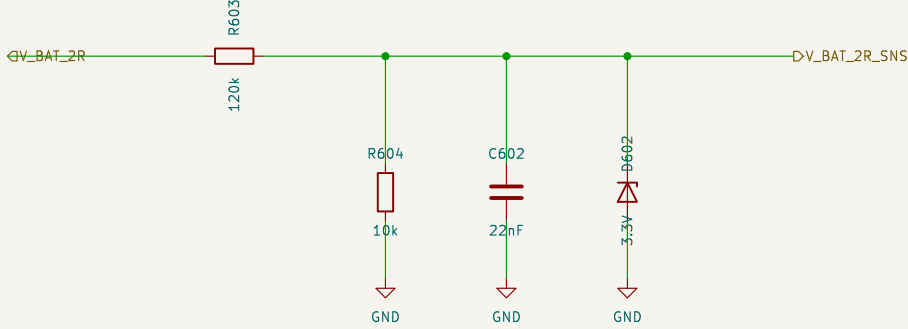
Title: Fuel metering unit and EGR actuator

Size: A2	Date:	Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/11

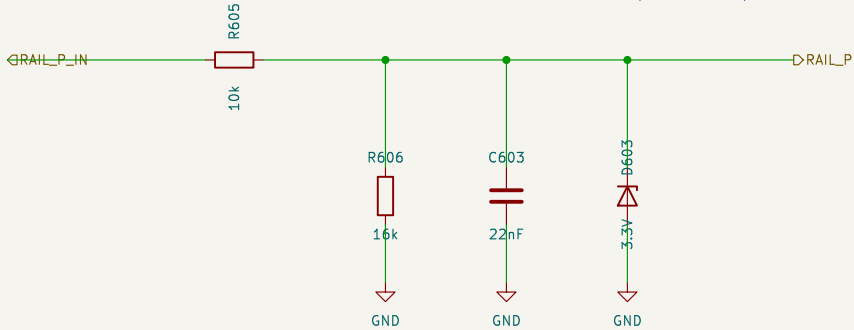
ECU pin 04 -- battery rail 1 -> PTA0 (ADC0_SE0)



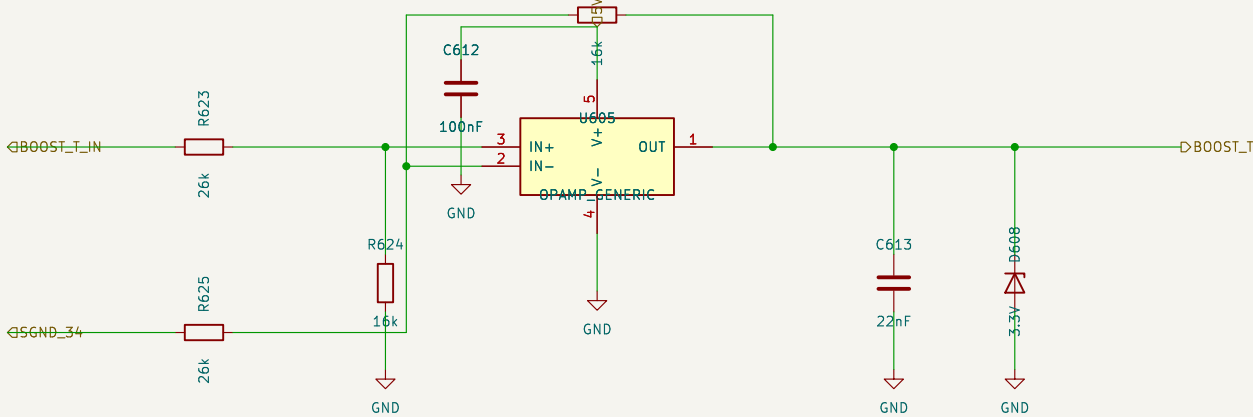
ECU pin 06 -- battery rail 2 -> PTA2 (ADC1_SE0)



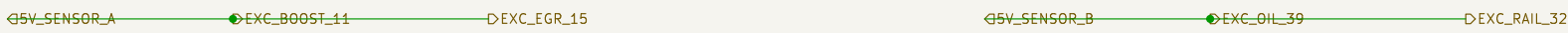
ECU pin 35 -- rail pressure, DEDICATED return pin 08 -> PTA16 (ADC1_SE13)



ECU pin 79 -- boost temperature -> PTA3. U2 OPEN: sensor type unmeasured



5V_SENSOR excitation -- three PTC-protected groups from the rails sheet, out to five connector pins



WHY FIVE CHANNELS ARE DIFFERENTIAL AND ONE IS NOT. The split follows the harness, not the sensor. Boost pressure and coolant temperature SHARE connector pin 34 as their return; EGR position and oil pressure share pin 36. Rail pressure has pin 08 to itself. A shared return carries both sensors' current, so its own resistance appears as an offset on both readings -- sensor_differential.cir puts 20 mA down a 1 ohm corroded return and measures a single-ended reading wrong by the whole offset while the differential one rejects it, a ratio of 90x. A dedicated return does not have that problem, so rail pressure gets the simpler front end.

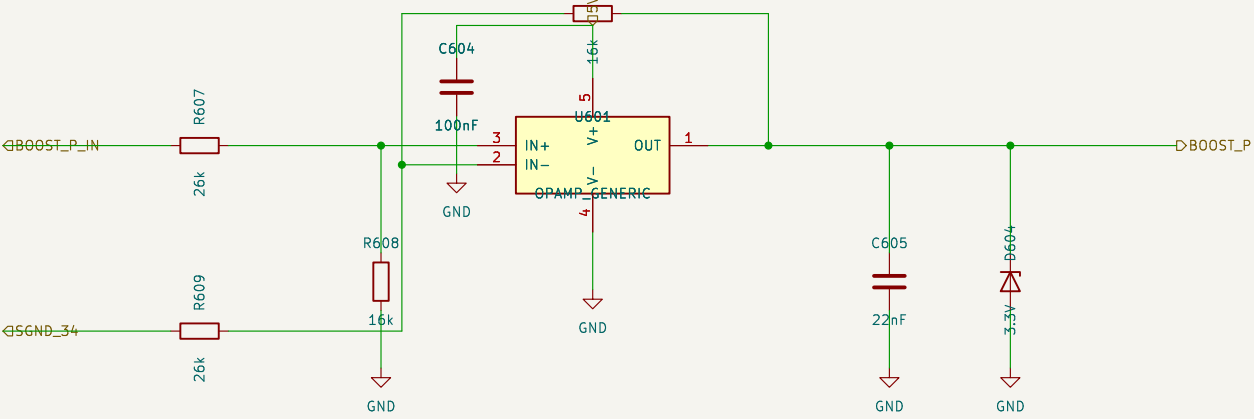
The block states the consequence as a harness requirement, which is the useful form: single-ended is only acceptable if contact resistance is maintained below a figure nobody is going to maintain over a genset's life in an Indian monsoon.

OP-AMPS: FUNCTION DRAWN, PART NOT CHOSEN. OPAMP_GENERIC has placeholder pin numbers and no footprint, deliberately -- placing a real symbol here would assert a choice nobody has made. The requirement IS specified; single 5 V supply, rail-to-rail input and output, and a difference network matched well enough for 60 dB CMRR, which is the floor sensor_differential.cir's own tolerance note tried and found the argument still holds at. The netlist models 80 dB and marks it INFERRED. 0.1% discrete resistors give roughly 48 dB and do NOT clear that floor. A matched network at 0.05% ratio, or a difference amplifier with the network on-die, does -- so this is a resistor-network decision as much as an amplifier one. Five channels is two quad packages.

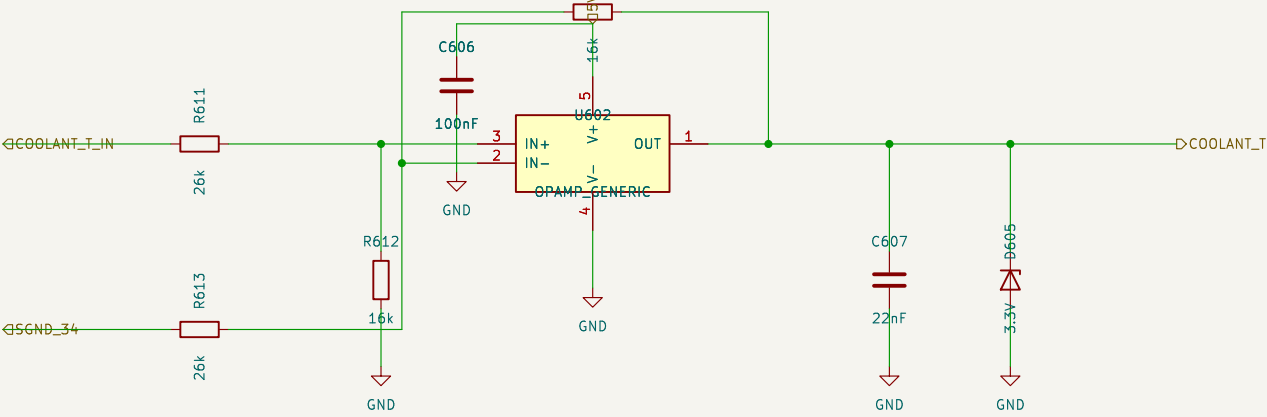
BOOST TEMPERATURE (PTA3, connector pin 79) is drawn differential like its neighbours, but U2 is still open: whether that element is a ratiometric sensor or a bare NTC is unmeasured. ntc_frontend.cir already chose between the two NTC options if it turns out to be one -- a 2.2k pull-up to 5 V, not a 100 uA current source, because the source needs 5.9 V of compliance at -40 C and the rail is 5 V. That is a populate-time change to this channel, not a redesign.

BARO (PTD18) is an onboard sensor, new per spec sec.7 deviation #2. No part is chosen, so nothing is drawn for it -- it needs a device with its own footprint, supply and output type, none of which follow from the pin assignment alone.

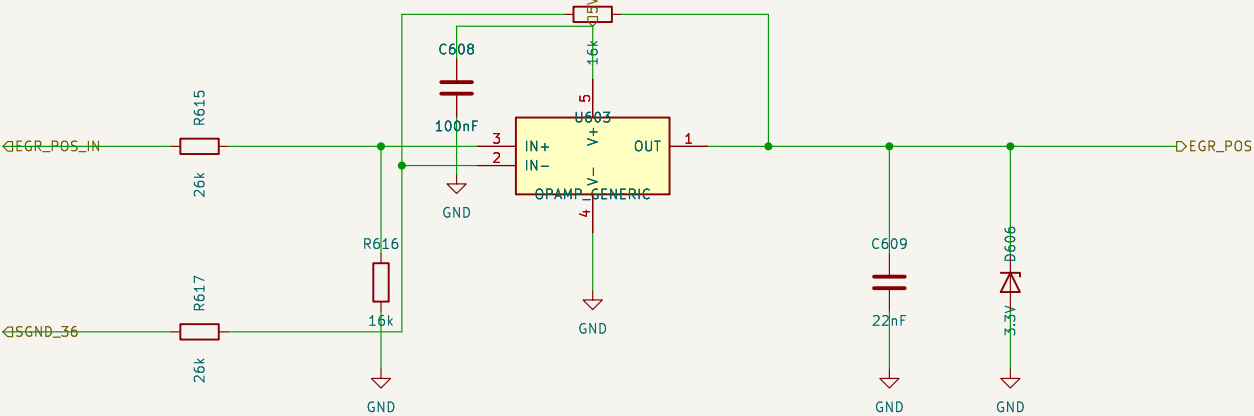
ECU pin 41 -- boost pressure, return SHARED on pin 34 -> PTA1



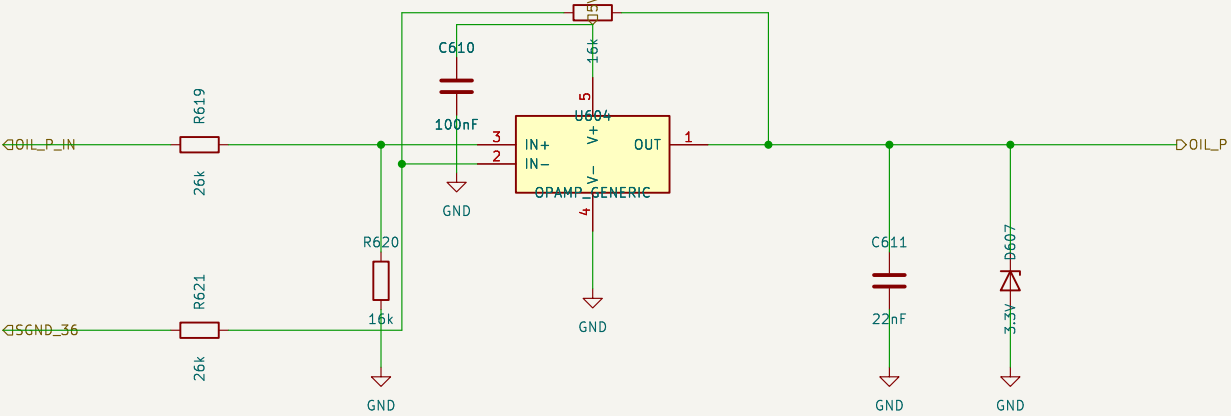
ECU pin 33 -- coolant temperature, return SHARED on pin 34 -> PTA6



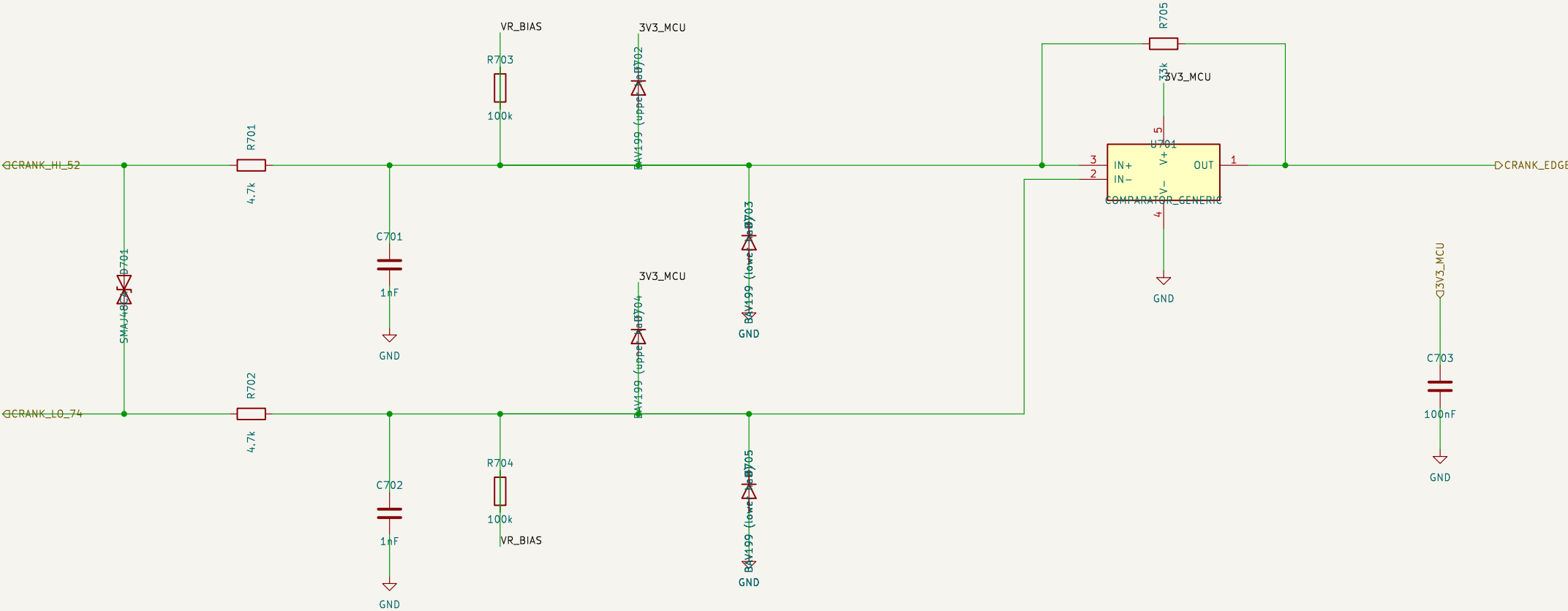
ECU pin 37 -- EGR position, return SHARED on pin 36 -> PTA15



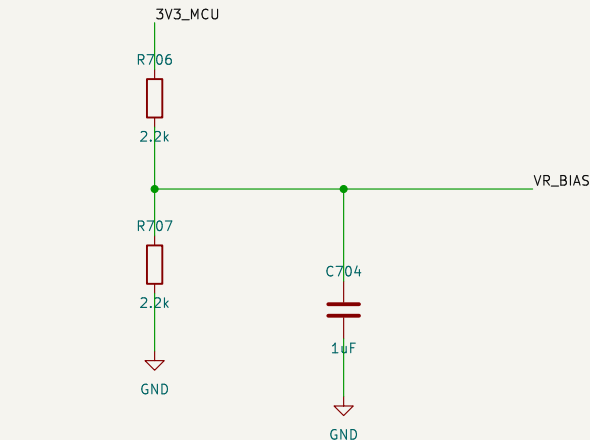
ECU pin 80 -- oil pressure, return SHARED on pin 36 -> PTA7



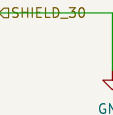
CRANK -- variable reluctance, ECU pins 52 / 74, shield 30 -> zero-cross comparator -> PTB2 (FTM1_CH0)



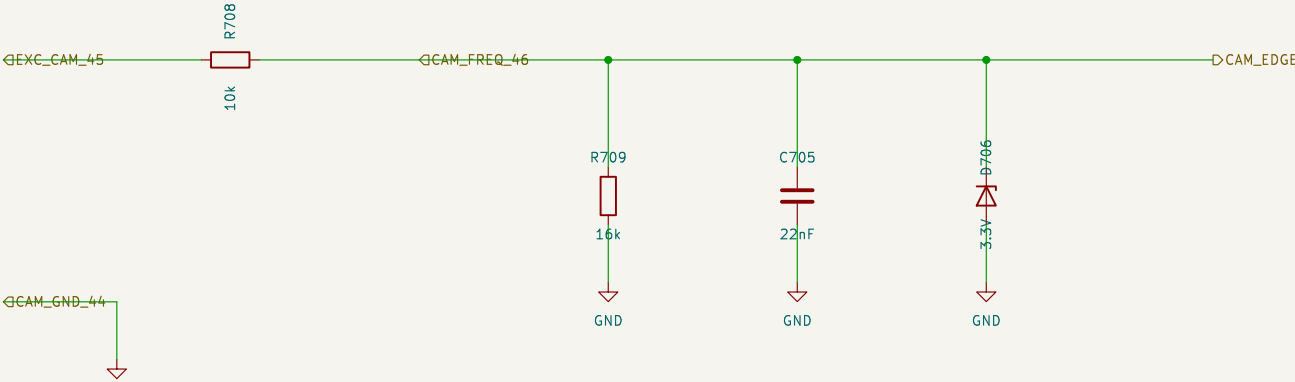
VR_BIAS -- mid-rail reference for the floating VR coil



ECU pin 30 -- crank sensor shield, terminated at THIS end only. Grounding both ends makes the shield a second conductor between two grounds that differ by the injector return current, and that current then flows down the shield around the pair it is there to protect.



CAM -- digital Hall, ECU pins 45 / 46 / 44 -> PTB3 (FTM1_CH1). CORRECTS the front-end spec sec.4 specifies literally



CLAMP LEAKAGE IS THE BINDING SPEC, WHICH INVERTS THE RULE THIS PROJECT USED ON THE POWER PATH. Every earlier clamp on this board was chosen for forward drop -- the negative-clamp Schottky's 0.547 V at 15 A decomposes into junction and bulk terms and that decomposition is what ruled out paralleling junction devices. Here the forward drop does not matter at all: these diodes conduct only on a fault, into a 4.7k series resistor that is already doing the limiting. What matters is what they do when they are OFF.

A reverse-biased diode's leakage flows out of the 4.489k Thevenin at the input node and appears as an offset voltage there. The hysteresis window is +/-198 mV, so:

BAT54S--class Schottky -2 uA at 25 C, and Schottky leakage rises roughly 100x to 125 C -> -200 uA -> 0.90 V of offset.
That is 4.5x the whole hysteresis window. The comparator latches and the engine has no speed signal at temperature.
BAV199--class low-leakage silicon 5 nA max at 25 C, under 1 uA at 125 C -> under 4.5 mV of offset, 2% of the window.

The two clamp diodes are biased symmetrically (both reverse-biased by 1.65 V from VR_BIAS) so their leakages partly cancel, and both legs are matched so what survives is partly common-mode -- but a design whose speed input depends on cancellation between two diodes' hot leakage is not one to defend. The requirement is stated instead: reverse leakage at or below 1 uA at 125 C. vr_conditioner.cir does NOT model this -- its comparator is an ideal switch with no input network at all, so no clamp, no leakage and no offset exist in it. This is the sheet finding something the block could not.

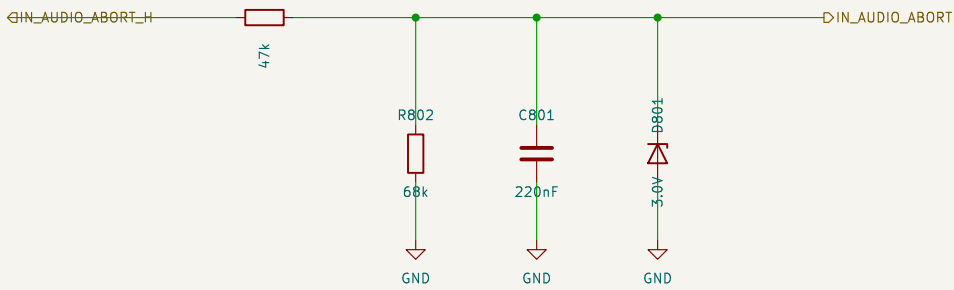
WHY THE CAM FRONT-END ON THIS SHEET IS NOT THE ONE THE SPEC ASKS FOR. Spec sec.4 and docs/pinmap.md sec.1.3 both give pin 46's front end as "Pull-up to 5V_SENSOR -> timer capture": the Hall output pulled to 5 V and taken straight into PTB3. NXP S32K1xx Data Sheet Rev. 15, Table 17, puts Vih max at VDD + 0.3 V, about 3.6 V at VDD = 3.3 V, so a full 5 V swing is not a guaranteed valid logic level. Table 1 does allow -0.8 to +5.8 V on any I/O pin, but only while respecting the +/-3 mA injection limit, and the datasheet's own words are that "functional operation at the maximum values is not guaranteed" -- a stress margin, not a design basis for a pin that would sit at 5 V once per cam revolution for the machine's running life. The 16k to ground turns it into the same 10k/16k divider the ratiometric sensor channels already use. It costs one resistor.

This was found by writing cam_frontend.cir, which pinmap sec.4 asked for by name before this sheet was drawn: no block had ever simulated a cam front-end -- vr_conditioner.cir covers the crank and there was no cam equivalent -- so a 5 V swing on a 3.3 V pin had never been checked. Fourth finding of the same shape in this project: a document and an executable file disagreeing, with nothing comparing them.

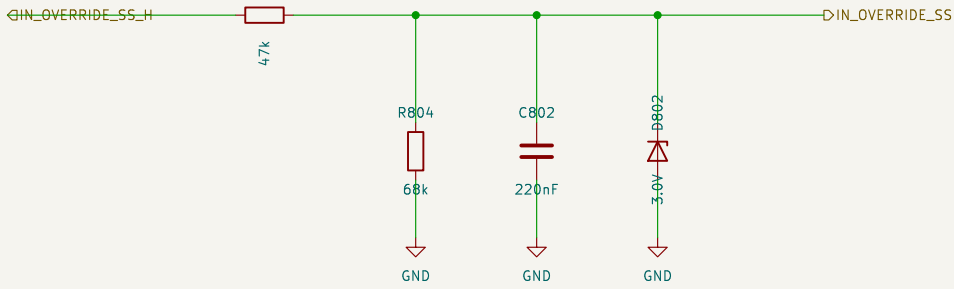
PARTS NOT CHOSEN ON THIS SHEET: the comparator (COMPARATOR_GENERIC -- placeholder pins, no footprint) and the cam Hall sensor itself, which is U8 on the unknowns register and needs the machine. The comparator's requirement is unusually easy, and deliberately so: biasing this VR_BIAS means the inputs never leave mid-rail, so no part is needed whose common-mode range includes ground -- the usual hard constraint on a zero-cross detector. Single 3.3 V supply, push-pull rail-to-rail output, propagation delay at or below 1 us (0.009 crank degrees at rated speed on a 60-tooth wheel). Input offset voltage is a non-issue against a 198 mV window.

Clamp diodes specified by REVERSE LEAKAGE, not forward drop -- see the sheet note		
Zero-cross with +/-198 mV hysteresis: a fixed threshold cannot see a 2 V cranking signal		
vr_conditioner.cir / cam_frontend.cir		
Generated by hw/gen_sheet_speed_inputs.py -- do not hand-edit until it is retired		
ecu25kva		
Sheet: /speed_inputs/		
File: speed_inputs.kicad_sch		
Title: Crank VR conditioner and cam front-end		
Size: A2	Date:	Rev:
KiCad E.D.A.	kiCad-cli 7.0.11+dfsg-1build4	Id: 8/11

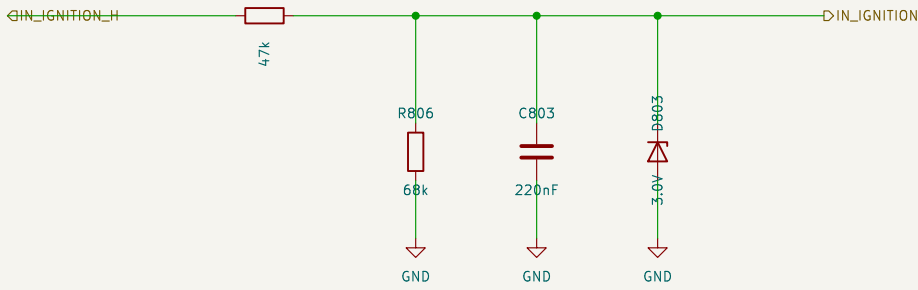
ECU pin 20 -- audio abort, active high -> PTD11



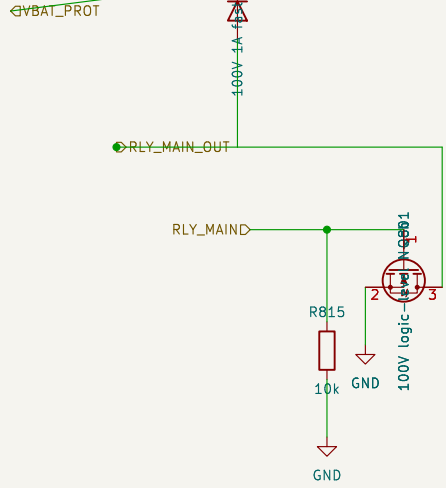
ECU pin 24 -- override start/stop, active high -> PTD12



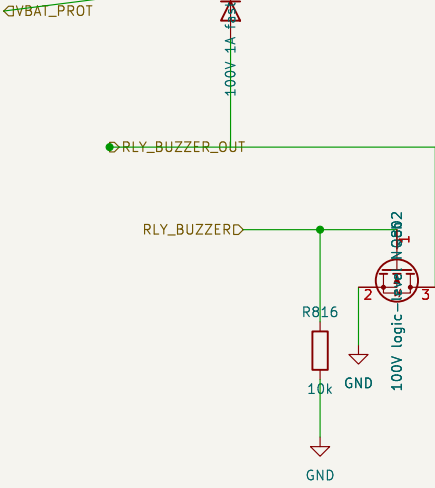
ECU pin 71 -- ignition, active high -> PTD16



ECU pin 50 -- main relay, low side <- PTD13



ECU pin 69 -- buzzer relay, low side <- PTD14

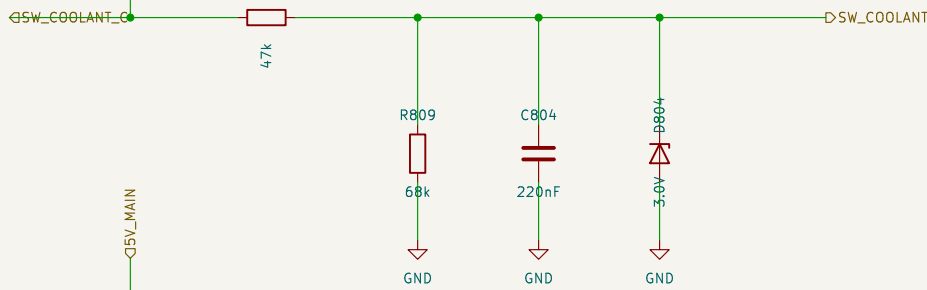


TRIP SENSE MOVES FROM PTD6 TO PTB0 (ADC0_SE4). docs/pinmap.md assigns it as a GPIO input; trip_module_sense.cir designs a THREE-STATE supervised loop and checks the three bands with real margins -- healthy 2.863-2.954 V, tripped 0.310-2.068 V, wire fault 0.000 V. A GPIO reads two states. It separates healthy from not-healthy and nothing else, which discards exactly the distinction the bleed resistor exists to create: a supervised loop that cannot report a cut wire is an unsupervised loop with extra parts. PTB0 is free, bonded out on the 144-LQFP, and Hi-Z with no pull at reset. PTD6 returns to the spare pool.

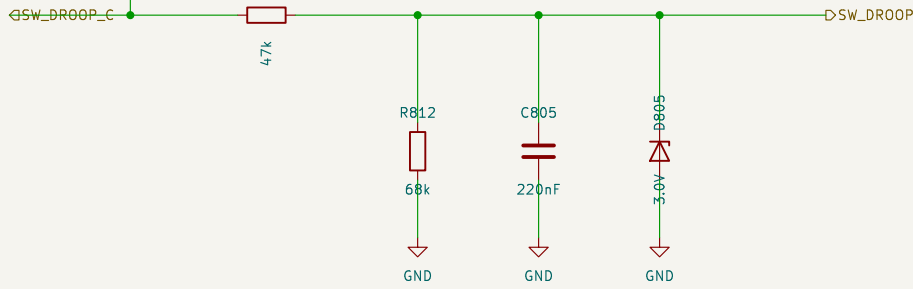
The 1.2 Mohm bleed resistor that makes the tripped band is INSIDE the trip module, across its contact -- it is not on this board and is not drawn here. It is also INFERRED rather than measured (trip_module_sense.cir says so), so the band edges above move if the real part differs.

WHY THE SAME FRONT-END ON EVERY CHANNEL. The 47k/68k/220nF/3.0V chain is discrete_input.cir's, and it is used unchanged on all six inputs -- including the two whose switch is a contact with real rather than a feed. Those two get a 10k pull-up BEFORE the 47k, so a battery short on the harness still meets the clamp chain rather than the pull-up. An open contact then leaves the divider at 2.72 V, under the 3.0 V knee, so it reads high without the zener conducting.

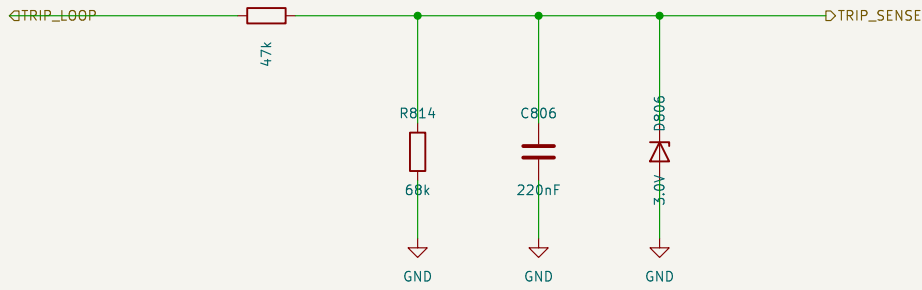
ECU pin 43 -- coolant switch, contact to ground -> PTD5



ECU pin 23 -- droop switch, contact to ground -> PTD7



NEW pin -- supervised trip loop -> PTB0 (ADC0_SE4), not PTD6



Relay FETs are 100 V class -- VBAT_PROT reaches 73.3 V on pulse 2a
Trip sense moved PTD6 -> PTB0: a GPIO cannot read a three-state loop
Front-ends: discrete_input.cir / trip_module_sense.cir. Outputs: relay_driver.cir
Generated by hw/gen_sheet_discrete_io.py -- do not hand-edit until it is retired

ecv25kva

Sheet: /discrete_io/
File: discrete_io.kicad_sch

Title: Discrete I/O: switch inputs, trip loop, relay drivers

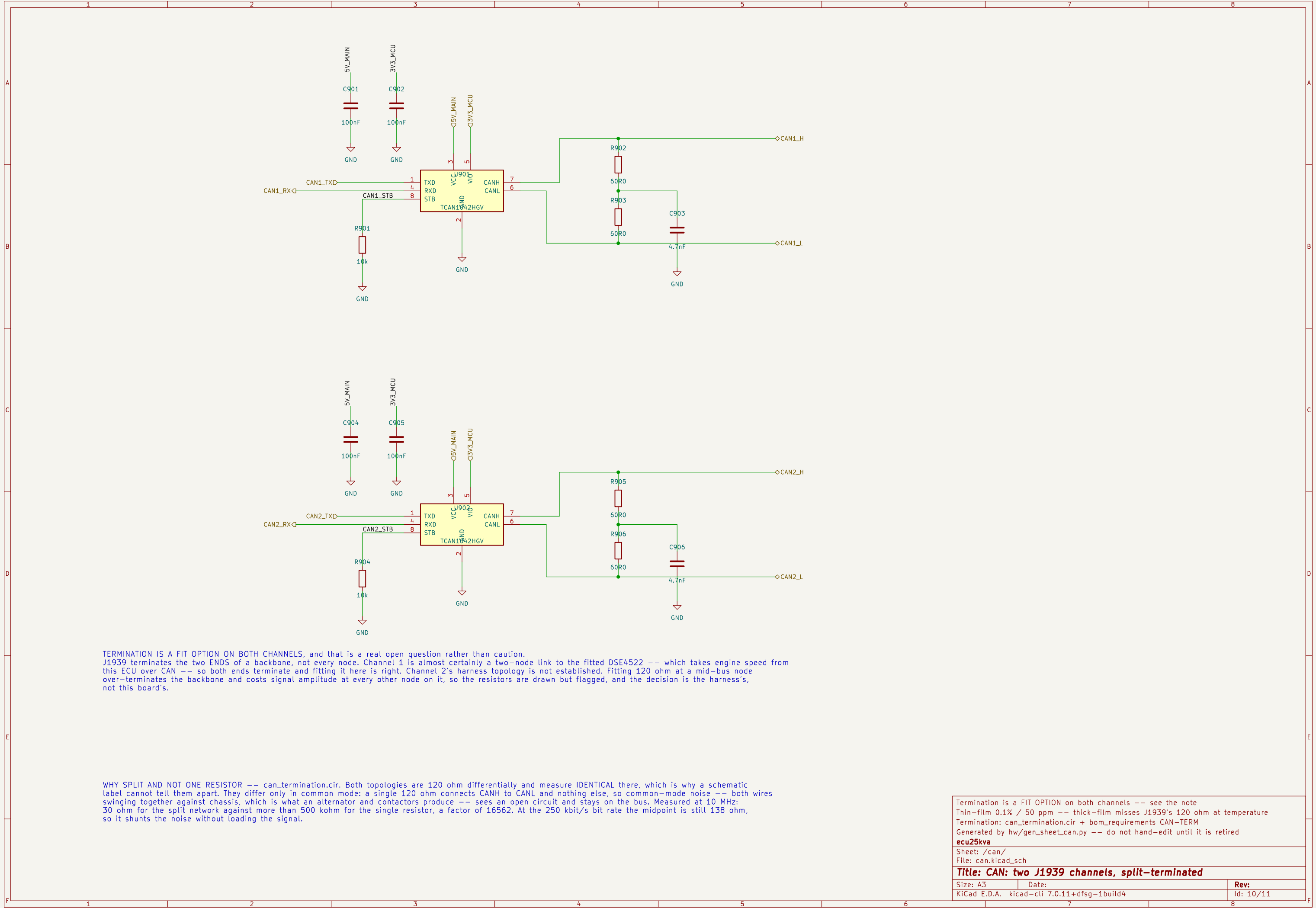
Size: A2

Date:

Rev:

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 9/11



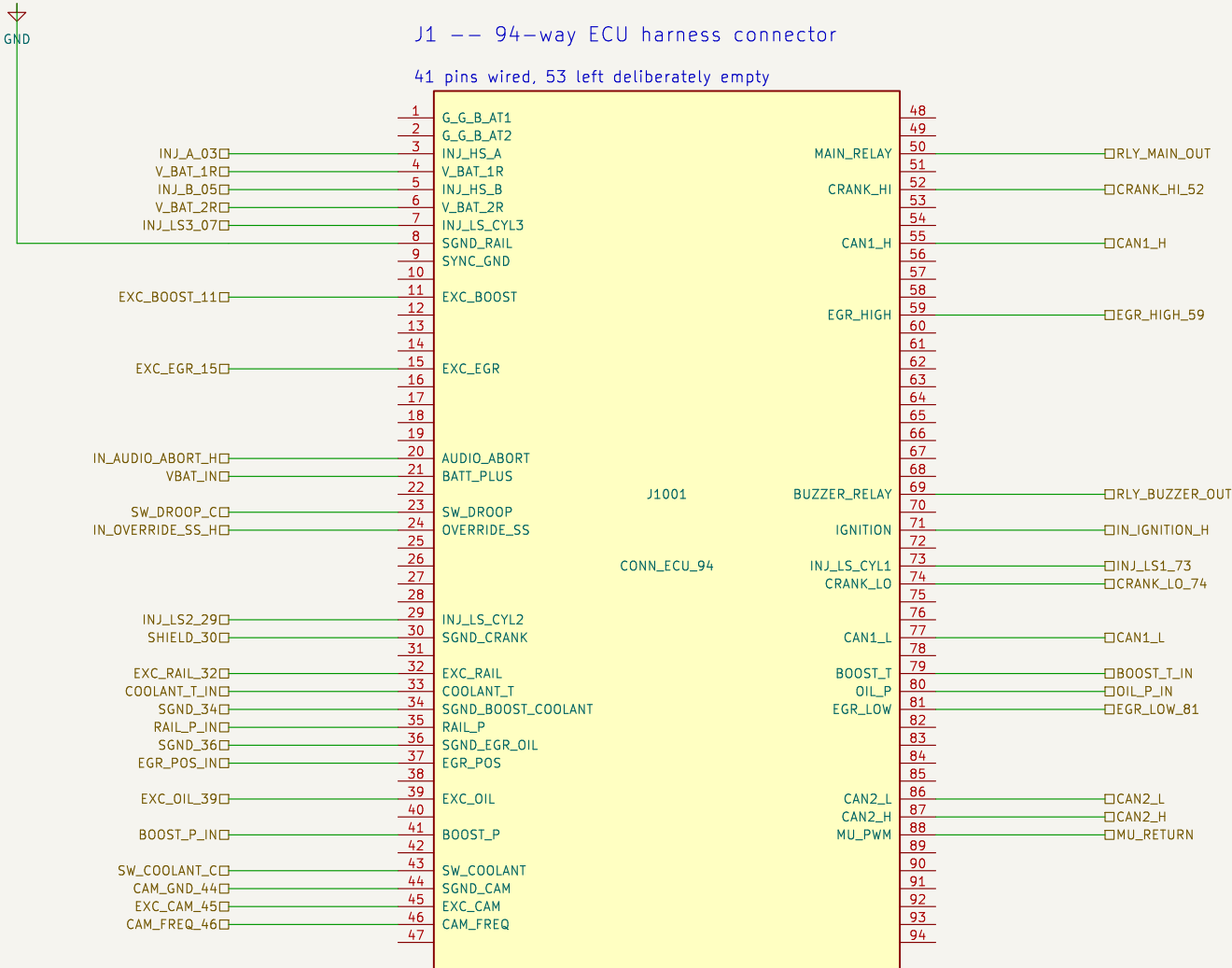
THE POWER GROUND HAS NO PIN, AND THAT IS THIS SHEET'S FINDING.

GP3.8703.C4.pdf shows pin 21 as the battery positive feed and shows no battery negative anywhere on the ECU connector. Its own scope note explains why -- It is a field troubleshooting diagram and it omits "power grounds, unused pins and internal-only pins" -- so this is a gap in the SOURCE, not a contradiction in it.

It is still a gap that stops a board. Every other sheet in this project returns to GND: the injector low sides carry 18 A into it, the metering unit up to 1.35 A, and the analog front ends reference their zeners and their 22 nf filters to it. At this boundary that net has nowhere to go.

It cannot be solved by choosing. Fifty pins have no established function and several of them are certainly grounds -- a 94-way connector feeding an 18 A injector stage does not have one return -- but picking which is exactly the guess this project forbids, and a wrong ground pin is not a rework, it is a harness that has to be remade.

WHAT ACTUALLY SETTLES IT: the connector part number and the OEM pin list, or continuity measurement on the machine between the connector shell and battery negative. The reference names the first as the missing document. The second needs the machine and belongs with U16/U2/U8/U14/U5.



PINS SHOWN ON THE DIAGRAM AND STILL NOT WIRED

- 09 SYNCHRONIZATION GROUND. Drawn RED on a diagram whose own legend makes red a positive feed, so the name and the colour disagree. U6 on the unknowns register.
- 01 G_G_B AT1. 02 G_G_B AT2. Memo 03 suspects these are power grounds rather than switch-return legs. If that is right they are part of the answer to the note on the left. No MCU pin is assigned either way, because guessing which would violate the same rule.

NETS THAT EXIST AND HAVE NO PIN YET

TRIP_LOOP the trip-module supervised loop is NEW -- it is not on the OEM diagram at all, so it needs one of the 50 unconfirmed pins. discrete_io draws the front end; the pin waits on the same document the ground does.

EGR_HIGH_59 wired here, because pins 59 and 81 ARE on the diagram -- but the bridge that drives them is the blocked half of metering_egr, waiting on a gate driver rated above the 73.3 V this board's own battery rail reaches on ISO 7637-2 pulse 2a. The pins are real; the driver is not chosen.

WHY PINS 34 AND 36 ARE SIGNALS AND PIN 08 IS GROUND

34 and 36 are SHARED sensor returns -- 34 for boost pressure and coolant temperature, 36 for EGR position and oil pressure. A shared return carries both sensors' current, so its own resistance is an offset on both readings, which is why sensors_analog carries them into difference amplifiers rather than to ground. 08 is rail pressure's DEDICATED return: nothing to reject, so it is simply ground here.

THE PART IS IDENTIFIED; THE LAND PATTERN IS NOT. refs/field-evidence-2026-09-18.md sec.2b closed U1 off a photograph of the housing angled to the light; BOSCH and the armature-in-circle logo moulded on the side body, 1 928 405 194 under the BOSCH text, 1 928 405 192 on the lower body strip, CODE C beside it, >PA66-GF50< on the locking lever. Read off the physical part, which beats a catalogue match because it cannot be a mis-identification. CODE C IS LOAD-BEARING: Bosch supplies these housings mechanically coded so differently-coded plugs cannot mate, so any replacement or adapter must match it or it will not engage -- the kind of detail that is invisible until a part arrives and will not fit. Still missing: the OEM PIN LIST, a different document from the part number, and the board-side footprint. A search for both numbers returned no catalogue hit, so the land pattern has to come through a Bosch distributor quoting them. Nothing is assigned here, because a mating land pattern is not a thing to infer from a photograph. Pin NAMES on the symbol are the wiring diagram's own signal names where it shows the pin and KiCad's blank elsewhere -- the same treatment the op-amps and the comparator get: the symbol asserts what is known and declines to assert what is not.

Bosch 1 928 405 192 / 194 code C -- part identified off the housing: land pattern is not			
50 pins have no established function -- including, so far, the power ground			
Pin count and 44 pin functions from GP3.8703.C4.pdf via refs/ecu-pinout-extracted.md			
Generated by hw/gen_sheet_connector.py -- do not hand-edit until it is retired			
ecu25kva			
Sheet: /connector/			
File: connector.kicad_sch			
Title: 94-way ECU connector and harness mapping			
Size: A2	Date:	Rev:	
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 11/11	